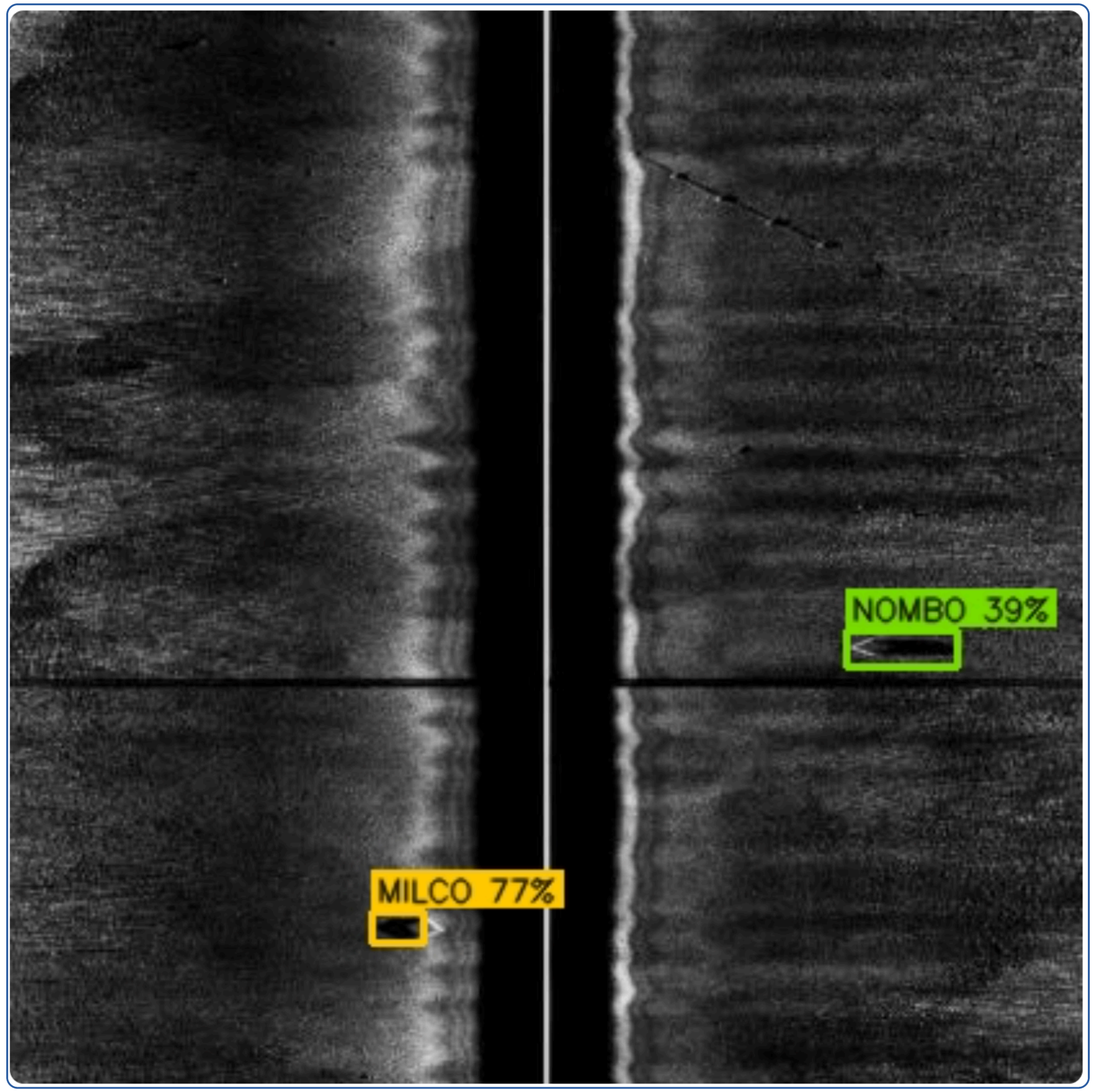
SMART INDIA HACKATHON 2026

- **Problem Statement ID – 26057**
- **Problem Statement Title** – AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery
- **Theme** – Disaster Management · Ministry of Earth Sciences (MoES) / NIOT
- **PS Category** – Software
- **Team ID** – Your Team ID
- **Team Name (Registered on portal)** – Your Team Name



Live output of our running prototype on a real held-out survey frame (`0460_2018`).
Yellow = man-made, green = ambiguous.

AQUA-SHIELD — Detect → Verify → Localize → Act. A working, offline side-scan sonar pipeline that does not just detect, but **verifies every detection against physical acoustic evidence** before an operator ever sees it.

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AQUA-SHIELD

Acoustic Intelligence for Underwater Anomaly, Debris & Marine-Hazard Localization

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Proposed Solution

- **Ten-stage offline pipeline.** Ingest → quality control → tiling → **YOLO11n detection** → **physical verification** → calibration → de-duplication → geolocation → priority → report.
- **Independent verification stage.** Every candidate is re-examined against **10 features measured from the pixels** — shadow coherence, local SNR, contrast, compactness, texture — that do *not* depend on the detector's own opinion.
- **Calibrated confidence.** Platt scaling fitted on a held-out survey; when unfitted, every hazard is stamped `calibrated: false`.
- **Ranked hazard register.** The output is not a model file — it is a prioritised, auditable list exported as `GeoJSON / CSV / SQLite`, opening directly in QGIS.
- **Runs fully offline on a laptop.** No cloud, no per-image cost, no data leaving the vessel.

THE TEN PHYSICAL FEATURES THE VERIFIER MEASURES

shadow ratio | shadow side consistency | local SNR | contrast | compactness | aspect ratio | edge density | relative texture | area fraction | range position

THE TEN STAGES — EACH SEPARATELY TESTABLE AND SEPARATELY ABLATED

Ingest › Quality control › Pre-process › Tiling › Detection › Verify › Calibrate › Dedup › Geo-locate › Report

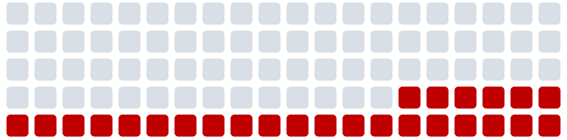
Teal = our core contribution. **Dashed amber** = built, measured, and **switched off by default because it made detection worse** — we keep negative results rather than hiding them.

Status: a working prototype, not a concept. It is trained, measured on held-out surveys, covered by 117 passing tests — and every screenshot in this deck is its live output.

How It Addresses the Problem

74% of frames in our held-out surveys contain **no target at all** (473 of 612 — measured). The real difficulty is not finding objects, it is **not crying wolf on empty seabed**.

- A recall-tuned detector alarms constantly, the analyst stops trusting it, and the system is abandoned. **Precision is the binding constraint** — so we built a stage specifically to enforce it.
- Replaces manual reading of thousands of km of sonar with a ranked register the operator can audit, accept or reject.
- Seabed clutter — rocks, sand ripples, acoustic shadows — is rejected on physical evidence, not on a hand-tuned threshold.



■ frames with a target
■ empty seabed
100 frames, to scale

Innovation & Uniqueness

- **Verification on independent physical evidence** — not a second neural network agreeing with the first.
- **Refusal as a feature.** No navigation metadata → no coordinate. We return `null` plus the reason. A fabricated latitude exports cleanly to CSV and sends a vessel to open water.
- **Confidence ≠ Priority.** "Is it real?" and "should you care?" are scored separately, so a low-confidence large hazard is not buried.
- **The verifier doubles as a diagnostic instrument.** Its fitted weights gave acoustic shadow a *negative* weight — the opposite of the physics. That exposed a real train/inference preprocessing mismatch in our own pipeline; fixing it moved F1 `0.012 → 0.144`. A hand-tuned threshold would have hidden the defect.

TECHNICAL APPROACH

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TECHNOLOGIES TO BE USED

Language & machine learning

Python 3.12 · PyTorch 2.13 · Ultralytics YOLO11n · scikit-learn · ONNX Runtime

Sonar & geospatial

OpenCV · NumPy · pyproj · GeoTIFF · GeoJSON → QGIS

Application & quality

Streamlit console · FastAPI REST · SQLite register · pytest · 117 tests

Compute targets

Apple MPS · CUDA · CPU fallback · Jetson-class edge (target)

KEY TECHNICAL DECISIONS — AND WHY

YOLO11n, not a transformer detector	2.58 M params fits edge hardware; reviewed DETR variants cost 2.6×–44× more for single-digit AP gains.
Logistic verifier, not a second CNN	Inspectable weights, per-detection attribution — an operator can see *why* a candidate was rejected.

DATA — REAL, LICENSED, LEAKAGE-FREE

DATASET	SCALE	SPLIT BY
MILCO / NOMBO CC BY 4.0	`465 / 93 / 612 frames` `191 test objects`	acquisition year
Derelict crab pot CC BY-SA 4.0	`6,674 images` `9,311 objects · 107 recordings`	recording ID

Never a random split. Consecutive sonar frames share seabed, gain settings and often the same object — a random split leaks and inflates every number. A unit test asserts the splits stay disjoint.

METHODOLOGY & PROCESS FOR IMPLEMENTATION

1 · ACQUISITION & CONDITIONING

Ingest — SSS frames + nav CSV / GeoTIFF › **Quality control** — dynamic range, speckle index, dropout, water column › **Preprocess** — OFF by default — measured 3 ways, it hurt every time › **Tiling** — 640 px overlap, seam merge by IoU / IoS

2 · DETECTION & VERIFICATION — THE CORE CONTRIBUTION

Detection — YOLO11n · 2.58 M params 6.3 GFLOPs › **10 physical features** — shadow ratio · shadow side · SNR · contrast · compactness · texture … › **Logistic verifier** — L2-regularised, fitted on a held-out survey › **Calibration** — Platt → probability, or stamped uncalibrated

3 · FUSION, LOCALIZATION & ACTION

De-duplication — repeat sightings → one unique hazard › **Geolocation** — GeoTIFF affine / per-ping nav **or refuse** › **Priority scoring** — separate from confidence › **Report** — ranked register → GeoJSON · CSV · SQLite

I/O CONTRACT `SSS frame 1024×1024 + nav › 640 px tiles › boxes + class + raw score › 10-feature vector → p(real) › hazard record › GeoJSON · CSV`

hazard record = `{id, class, level1, confidence, calibrated, priority, priority_band, lat, lon, uncertainty_m, observations, evidence[10], provenance}` — every field an operator needs to act on, audit, or reject the call.

WORKING PROTOTYPE — THE OPERATOR CONSOLE

Frames	Raw candidates	Filtered out	Unique hazards	High priority	ms / frame
8	2	0 ↓ -0%	2	1	27

Run summary: 8 frames → 2 raw candidates → 2 unique hazards → 1 HIGH priority.

hazard_id	class	level1	confidence_%	band	calibrated	priority	priority_band	obs	lat	lon	±m	length_m	frames
AQS-00001	mine_like_object	MAN_MADE	76.99	HIGH	✓	58.7	HIGH	1	None	None	None	None	0460_2
AQS-00002	bottom_object_uncertain	AMBIGUOUS	39.48	LOW	✓	33.2	ROUTINE	1	None	None	None	None	0460_2

The hazard register. This survey ships no navigation, so `lat / lon / ±m` come back None — the system **refuses to invent a coordinate** rather than guessing.

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FEASIBILITY AND VIABILITY

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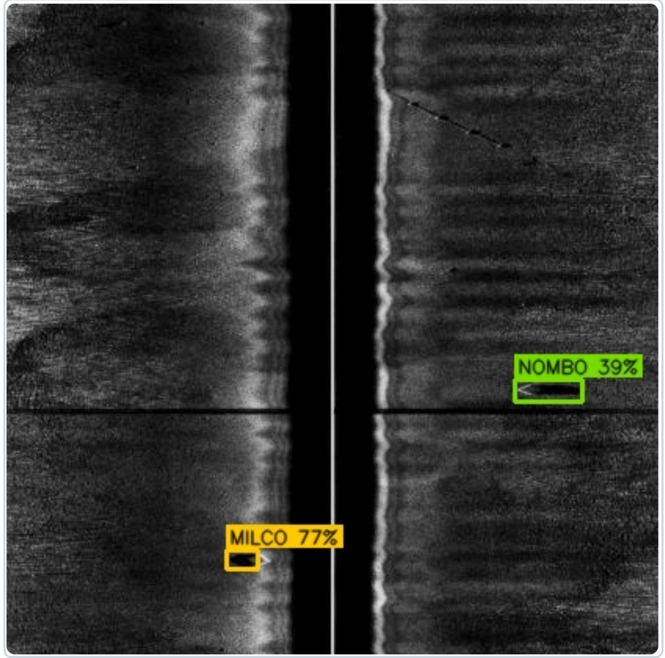
Analysis of the Feasibility

It already runs. A live run on 8 held-out frames: 8 frames › 2 raw candidates › verify › 2 unique hazards › 1 HIGH priority, at 27 ms per frame.



Live prototype output, frame `0460_2018` — not a mockup.

MEASURED ON THIS MACHINE	
MPS inference	`21.4 ms`
ONNX export	`10.6 MB · 8.5 ms`
Throughput	`37.4 frames/s`
Peak memory	`640 MB`
Test suite	`117 passing`

Apple M5, 24 GB — a laptop, not a cluster. Reproducible from `experiments/`.

VERIFICATION STAGE — MEASURED EFFECT (612 HELD-OUT FRAMES, 473 EMPTY)

CONFIGURATION	PRECISION	TRUE POSITIVES	FALSELY-ALARMED FRAMES
Detector only	0.247	21	37 / 473
+ hand-written rules	0.300	12	18 / 473
+ learned verification	**0.322**	**19**	**25 / 473**

Hand-written rules buy quiet by discarding targets (21 → 12). The learned verifier keeps **19 of 21** while cutting falsely-alarmed frames by **32%**. That gap is the case for fitting the filter rather than tuning thresholds.

TARGET VALIDATION — WHAT WE WILL PROVE AT THE FINALS

METRIC THAT MATTERS	MEASURED TODAY	TARGET
False alarms on target-free frames	`25 / 473 (5.3%)`	`≤ 2%`
Precision @ IoU 0.3	`0.322`	`≥ 0.60`
Recall @ IoU 0.3	`0.099`	`≥ 0.45`
Ghost-gear detection, mAP50	`0.323 raw, 1 run`	`≥ 0.55`
Calibration error (ECE)	fitted, not yet reported	`≤ 0.05`
Geolocation error (CEP)	not measurable — no nav in available data	`≤ 10 m`
Inference latency / frame	`21.4 ms MPS`	`≤ 30 ms edge`

Measured = reproducible from `experiments/` on held-out surveys. **Target** = a goal, not a result. We publish both columns so the gap is visible rather than hidden. **Baselines we will measure against:** detector-only · hand-written rule filter · YOLO11s / RT-DETR at equal compute.

POTENTIAL CHALLENGES & RISKS → STRATEGIES FOR OVERCOMING THEM

CHALLENGE / RISK	STRATEGY — ALREADY BUILT
Scarce labelled coastal SSS data	Leakage-free recording-level splits; two licensed datasets; honest reporting instead of inflated splits.
Detection degrades under speckle noise	Noise-robust checkpoint trained and shipped separately, labelled as a trade-off rather than silently swapped in.
Detector backend is AGPL-3.0	Isolated behind an interface; the ONNX export path is backend-independent.
Edge compute is constrained on an AUV	10.6 MB ONNX model measured at 8.5 ms — sized for Jetson-class hardware.
Missing navigation metadata	The system refuses to emit a coordinate and states why, rather than guessing.

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IMPACT AND BENEFITS

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Potential Impact on the Target Audience

TODAY
analyst scrolls raw sonar imagery

›

WITH AQUA-SHIELD
ranked hazard register, evidence attached

›

ACTION
GeoJSON → QGIS, vessel tasked

BENEFICIARY	IMPACT
MoES / NIOT survey & cleanup operations	Direct fit to the problem statement's own agency; runs offline on survey hardware already aboard.
Fisheries & coastal management	Ghost gear keeps killing catch and habitat long after it is lost; early detection breaks that cycle.
Ports & navigation safety	Mine-like and man-made bottom objects flagged and prioritised before they foul a channel.
Environmental researchers	Auditable, licence-clean datasets and a reproducible experiment registry.

DEPLOYMENT & SCALABILITY — OFFLINE-FIRST BY DESIGN

- **Today:** runs on a single laptop (Apple M5, 24 GB), fully offline.
- **Edge:** 10.6 MB ONNX export measured at 8.5 ms — sized for a Jetson aboard an AUV or survey launch.
- **Fleet:** FastAPI + SQLite register scales the same pipeline from one operator console to a shore-side survey fleet.
- **Not yet:** never run on a Jetson or a live AUV. That is the next milestone, and we state it as a target rather than an achievement.

Benefits of the Solution

DIMENSION	BENEFIT
Environmental	Faster location of ghost nets and derelict gear that keep entangling marine life and smothering reefs after they are lost.
Economic	No cloud inference, no per-image API fee, no data egress — the marginal cost of one more survey is the electricity to run a laptop already aboard.
Operational	32% fewer falsely-alarmed frames while keeping 19 of 21 true positives (measured) — the analyst reviews a ranked register instead of raw imagery.
Safety	Mine-like and man-made hazards prioritised before a vessel or diver encounters them.
Social & institutional	Every hazard carries evidence, calibrated confidence and provenance, so a decision can be audited — not merely trusted.

THE EXPORTED HAZARD REGISTER — FROM THE RUNNING SYSTEM

hazard_id	lat	lon	class	confidence_pct	priority	priority_band	uncertainty_m	observations
AQS-00001	12.9208	80.3399	mine_like_object	39.08	53.9	ELEVATED	6.01	1
AQS-00002	12.9206	80.3399	mine_like_object	37.06	54.4	ELEVATED	5.99	1
AQS-00003	12.9199	80.3404	mine_like_object	61.69	60.8	HIGH	6	1
AQS-00004	12.9207	80.3406	mine_like_object	37.57	53	ELEVATED	5.93	1

Real coordinates, per-hazard uncertainty in metres, and a priority band — the artefact a cleanup vessel is tasked from. **The navigation track in this demo is synthetic and the product says so on screen:** the geolocation path works end to end; the fix is not validated.

What we deliberately do not claim. No "% of analyst time saved" appears on this slide — that needs a user study we have not run.

−32%
falsely-alarmed frames
measured

19/21
true positives kept
measured

37.4/s
frames processed
measured

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RESEARCH AND REFERENCES

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DATASETS & PRIOR ART — EVERY LICENCE VERIFIED

WORK	LICENCE	RELATIONSHIP
MILCO / NOMBO Data in Brief 53:110132 · DOI 10.6084/m9.figshare.24574879	**CC BY 4.0**	**Our training data** — mine-like & bottom objects.
sss-crab-pot-detection-ds PINGEcosystem · HuggingFace	**CC BY-SA 4.0**	**Our training data** — derelict ghost gear.
GhostVision JMSE 14(10):951, 2025	NOASSERTION	Closest prior system. Not vendored — licence unresolved.
PINGMapper Earth & Space Science, 2022	MIT	Pipeline shape and output conventions.
sidescantools	GPL-3.0	Studied; not vendored.
AI4Shipwrecks arXiv 2401.14546	MIT	Route to a wreck class later.

METHOD REFERENCES

- **Lee (1980)** — adaptive speckle filtering, DOI 10.1109/TPAMI.1980.4766994
- **Platt (1999)** — probabilistic outputs for large-margin classifiers, MIT Press
- **Guo et al. (2017)** — on calibration of modern neural networks, arXiv:1706.04599
- **Ultralytics YOLO11** — detection backbone (AGPL-3.0, interface-isolated)

ARCHITECTURES REVIEWED AND DELIBERATELY NOT ADOPTED

SSM-DETR · TR-YOLOv5s · MSF-DETR · LEF-RT-DETR. Each needs **2.6×–44× our compute** for single-digit AP gains, measured on forward-looking sonar or on datasets we cannot obtain to reproduce. Our bottleneck is false alarms on empty seabed — a bigger backbone does not fix that.

HOW OUR APPROACH COMPARES

CAPABILITY	AQUA-SHIELD	MANUAL ANALYST	GENERIC YOLO	PUBLISHED SSS DETECTORS
Detects bottom objects in SSS	✓	✓	✓	✓
Independent physical verification	✓	✓	✗	✗
Calibrated confidence	✓	✗	✗	rare
Refuses to invent a coordinate	✓	✓	✗	✗
Per-detection evidence & audit trail	✓	informal	✗	✗
Runs offline on survey hardware	✓	✓	✓	varies
Scales to thousands of km	✓	✗	✓	✓

WHAT WE HAVE NOT DONE — STATED DELIBERATELY

- Ghost-gear detection has **one raw training result** (mAP50 0.323). It has not yet been through the verification stage, calibration, or an ablation.
- Geolocation accuracy has **never been validated** — the licensed data available to us ships no navigation track.
- Detection **degrades under added speckle noise** (measured).
- Never run on a **Jetson or a live AUV**.
- Our own sonar preprocessing chain **did not help** when measured properly, and ships disabled by default.

Roadmap to the finals: 1 · fit + ablate verification on the ghost-gear model · 2 · run the baseline comparison · 3 · report the reliability curve and ECE · 4 · benchmark the ONNX export on Jetson · 5 · validate geolocation against a survey that ships a navigation track.